A THIRD-PARTY RE-EXECUTION OF THE EXACT SIGN CERTIFICATE FOR THE FERROMAGNETIC FOUR-STATE POTTS RECONSTRUCTION THRESHOLD

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Abstract. The OpenAI Math Release preprint *The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model* proves nonreconstruction at and below the Kesten–Stigum bound $d\lambda^2 \le 1$. Its proof is computer assisted: via Proposition 3.1 and Lemma 6.1 it rests on two exact-arithmetic subdivision traversals (Proposition 6.2), whose counts (Table 2) the authors obtained with GCC 13.3.0. The bundled lightweight checker validates the recorded logs but does not repeat the traversals. We re-ran both full traversals on a different machine with another GCC version (9.5.0) and Boost 1.86, and obtained stderr traces that are byte-identical to the published ones. We also re-ran traversal (a) in full with Clang 17.0.6 (`-O2`) and GCC 12.3.0 (`-O1`), again byte-identical, and the first part of traversal (b) with Clang. This is a reproducibility record. It contains no new mathematics, and it does not check the reduction from the theorem to the coefficient arrays.

Attribution. The theorem, the reduction and the verifier `verification/checks/certificate.cpp` are OpenAI's work (family 229 of https://github.com/openai/math, commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, preprint [1]). We only executed the published program and compared its output. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; an independent adversarial review was run by a separate AI session and is not human peer review.

1. What the certificate is

For the channel $P_\lambda(j \mid i) = \lambda \mathbf{1}_{i=j} + (1-\lambda)/4$ on [4], [1, Theorem 1.1] states that $a_n(d,\lambda) \to 0$ whenever $d\lambda^2 \le 1$, on d-ary and Poisson(d) trees. The probabilistic argument reduces this to two polynomial inequalities on a product of simplices [1, Proposition 3.1]. These are certified by Bernstein-type coefficient signs on a midpoint subdivision [1, Lemma 6.1, Proposition 6.2]. The verifier is a single C++17 file (185 lines, exact integer and `boost::multiprecision::cpp_int` arithmetic, `assert` enabled). It prints one progress line every 100 visited cells (mode, cell count, current depth, Type P and Type T leaves so far) and finally

`done 4479 19 lp=2074 lt=166` (row a), `done 1935 20 lp=871 lt=97` (row b),

which are the counts of [1, Table 2]. The trace samples cumulative counts every 100 cells of a deterministic depth-first traversal, so equality of full traces is strong evidence (though not a proof) that the same cells were visited and accepted in the same order.

2. What was already on record

The upstream file `checks/full-runs.json` records the authors' runs: GCC 13.3.0 (Ubuntu 13.3.0-6ubuntu2~24.04.1), flags `-std=c++17 -O3`, elapsed 98.1 s and 1287.9 s, with log hashes. The bundle also contains `mode-a-ubsan.log`, identical to `mode-a.log` and not referenced anywhere, apparently a second run of (a) under UBSan. The lightweight checker `checks/verify.py` rebuilds small parts under UBSan, checks the source binding and the recorded logs, and prints *"Full traversals were not repeated"*. So the counts rest on the authors' own executions. We found no third-party

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record of a full traversal; a community project (`cert-machine`) lists this preprint as not yet checked and plans a clean-room re-implementation.

3. What we ran

Machine: Intel Xeon Gold 6230R, Linux 6.8 (Ubuntu 22.04.5), single thread per run, Boost 1.86 headers. Source: `certificate.cpp`, SHA-256

`9f9765fb4dee9f2fc618390df89d3abc f2602569709fd4c2133a201b6e86bff`

(equal to the hash in `full-runs.json`). All builds use `-std=c++17 -UNDEBUG`.

Date	Compiler	Opt.	Run	Result (user time)
2026-10-07	GCC 9.5.0	`-O3`	(a) full	trace identical (5m51s)
2026-10-07	GCC 9.5.0	`-O3`	(b) full	trace identical (45m06s)
2026-10-09	Clang 17.0.6	`-O2`	(a) full	trace identical (2m54s)
2026-10-09	GCC 12.3.0	`-O1`	(a) full	trace identical (4m00s)
2026-10-09	Clang 17.0.6	`-O2`	(b) 30 min	8-line prefix identical (19m40s)

"Trace identical" means that the complete standard-error output is byte-for-byte equal to the upstream `checks/logs/mode-a.log` (resp. `mode-b.log`). These files are equal to `paper-mode-a.log` (resp. `paper-mode-b.log`) with the authors' trailing `time` output removed. The 2026-10-07 binary was identified afterwards: rebuilding with `g++-std=c++17-O3-UNDEBUGchecks/certificate.cpp` in the same directory reproduces its SHA-256 exactly. The helper `small_polynomials.py` (the four scalar minima of [1, Lemma 5.1] and the 24 coordinate permutations of [1, Section 5]) gives JSON equal to the recorded file `paper-small-polynomials.json`.

The full traversal (b) was not repeated with a second compiler, because it takes more than 30 minutes on this machine. The script `scripts/run_all.sh` reruns the hash checks, the trace comparisons, a fresh full traversal (a) with a compiler of the user's choice, and optionally (b) in full or as a time-limited prefix.

4. Scope

This record shows that the published program, run as published, produces the published counts on an unrelated toolchain and machine, and at three optimization levels. It does not check that the program computes what [1, Sections 3–6 and Appendix A] say it computes. It also does not check the reduction from Theorem 1.1 to the coefficient arrays. Nor is it an independent re-implementation of the subdivision. Because the arithmetic is exact, a compiler-independent trace is the expected outcome. Its value is evidence against undefined behaviour that changes results on the toolchains tested, and against errors in the published logs. The archived traces contain no host or compiler data; our local file times and `time` output support real runs, but a reader can re-check only by re-running (mode (a) takes a few CPU minutes, mode (b) 20–45).

References

[1] OpenAI, *The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model*, OpenAI Math Release preprint, October 5, 2026, https://github.com/openai/math/tree/main/preprints/The-Reconstruction-Threshold-for-the-Ferromagnetic-Four-State-Potts-Model-October-5-2026.